

Linux Privilege Levels: Kernel Mode and User Mode

The level of access to system resources and functions is determined by the two execution modes that Linux uses: Kernel Mode and User Mode.

1. Overview of Privilege Level

When running in kernel mode, also referred to as supervisor mode, the kernel has total access to all system resources including memory and hardware.

In the User Mode, User applications run in this mode with restricted access to system resources to avoid harm, interference with other applications.

2. Full Rights in Kernel Mode

The kernel has unrestricted access to memory, hardware, process, and file systems.

kernel operations include memory management, hardware access, process management, interrupt handling, system calls and network management.

3. User Mode: Limited Access

User apps cannot directly communicate with hardware or other essential system components and have limited access.

Examples of user operations include network requests, file operations, memory allocations (via system calls) and active user software.

4. Changing Context: A context switch or the transition from kernel mode to User mode, is brought on by things like system calls, interrupts and exceptions.

5. Dual-Mode Operational Security & Stability Benefits:

Prevents malicious software and system crashes.

Fault isolation ensures that when one application fails, it does not affect the system as a whole.

Resource Management: To ensure system stability, the kernel distributes system resources equitably.